PROPWISE AI

Model Evaluation Report

250,000 listings · 6 classifiers · 5 regression baselines · generated 28 Oct 2026 12:43 UTC

Key results

- Champion classifier: XGBoost on the listing track, test ROC-AUC 0.9991, F1 0.9839, ECE 0.0031. It reproduces a transparent rule (₹/sqft vs. comparable median), so the near-perfect score is expected and is not evidence of predicting returns.
- With price removed, the best model (Logistic Regression) reaches ROC-AUC 0.7825, almost entirely through area.
- Regression on current price finds no signal: best learned test R^2 -0.0002. The app therefore shows no 'estimated value' and uses explicit growth scenarios for five-year figures.
- All numbers are read from models/evaluation.json written by scripts/train_models.py and cross-checked against the MLflow tracking store; none are typed by hand.

1. Setup

LISTINGS	TRAIN / VAL / TEST	GOOD-VALUE SHARE	COHORTS
250,000	**70% / 15% / 15%**	**50.0%**	**630**
SHA-256 b0032196fcf3...	37,500 test rows · seed 42	test (train 50.1%)	smallest 165 train rows

Label and selection

Label. Rupees per sqft at or below 1 x the training-split median of its City x Property_Type x BHK cohort (fallback City, then global, when a cohort has < 30 rows).

Selection. Highest validation roc-auc on the listing track; ties within 0.001 broken by lower validation brier score. Test metrics were not used for selection.

Validation design. Rows are independent listings with no repeats and no time index, so a random stratified split is appropriate; group- or time-aware validation is not required.

2. Investment classification

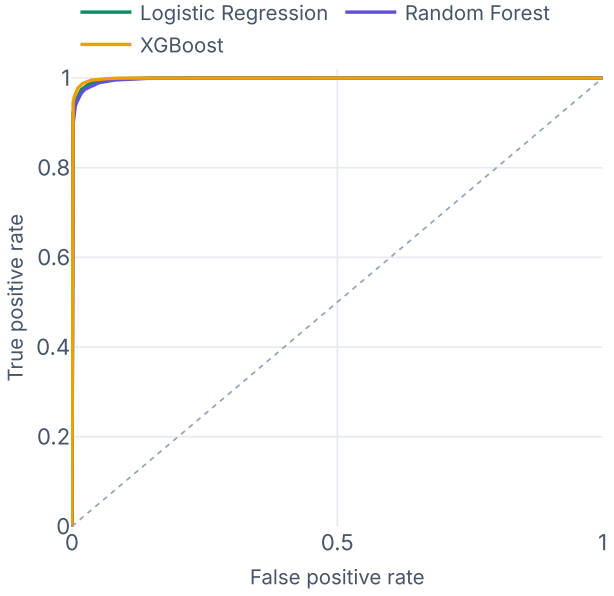
Metrics (test split unless marked)

Track	Model	Val AUC	Test AUC	Precision	Recall	F1	Accuracy	Brier	ECE
listing	Logistic Regression	0.9984	0.9985	0.9783	0.9775	0.9779	0.9779	0.0154	0.0028
listing	Random Forest	0.9976	0.9978	0.9761	0.9747	0.9754	0.9754	0.0279	0.0792
listing	**XGBoost ★**	**0.9990**	**0.9991**	**0.9832**	**0.9845**	**0.9839**	**0.9838**	**0.0116**	**0.0031**
attributes only	Logistic Regression	0.7804	0.7825	0.7069	0.7238	0.7152	0.7115	0.1905	0.0154
attributes only	Random Forest	0.7794	0.7815	0.7018	0.7350	0.7180	0.7111	0.1913	0.0196
attributes only	XGBoost	0.7752	0.7775	0.7036	0.7188	0.7111	0.7077	0.1924	0.0114

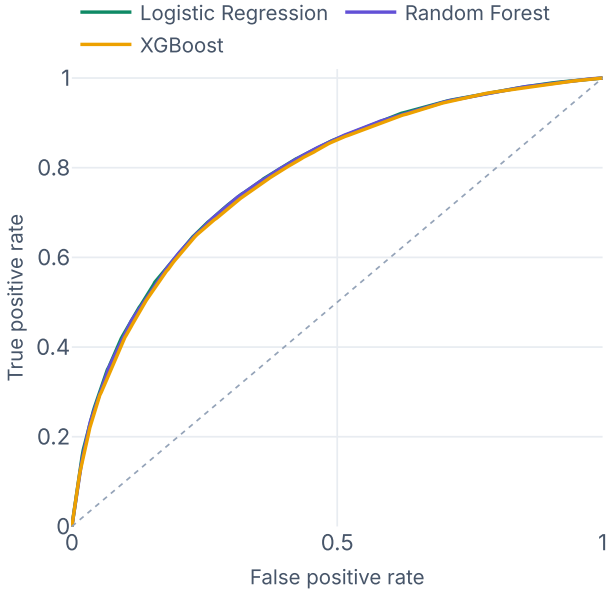
★ = champion (highlighted). **Listing** track includes price and area; the label is a deterministic function of price ÷ area against a cohort median, so these models learn to reproduce the rule. **Attributes only** removes price; its signal comes almost entirely from area.

ROC curves

Listing track (price included)

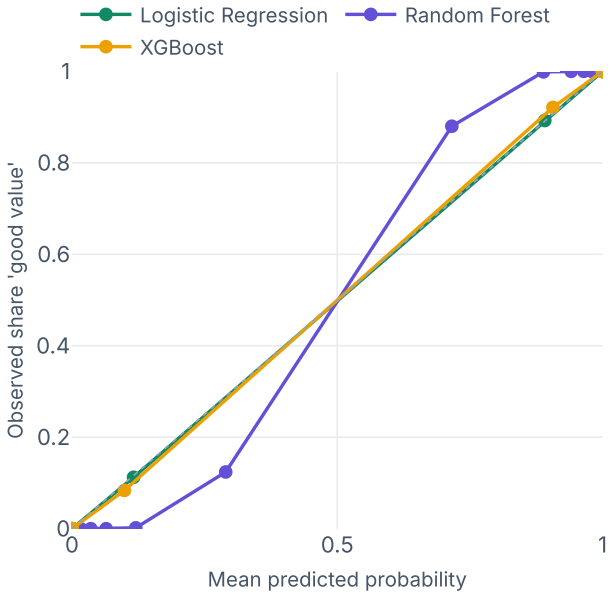


Attributes only (price removed)

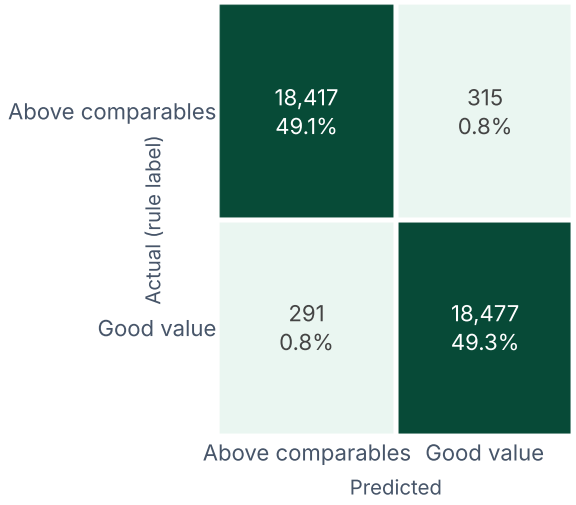


Calibration and errors

Calibration: listing track



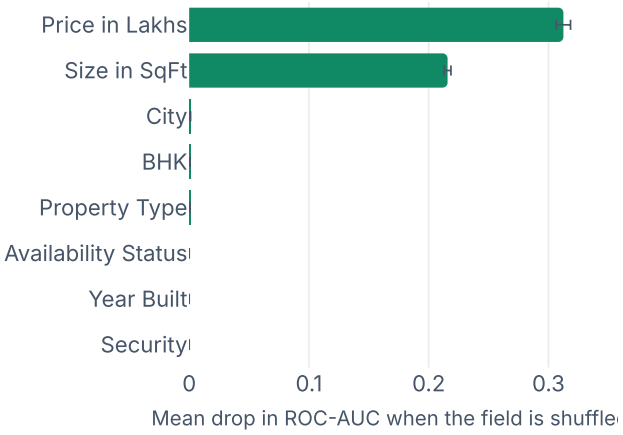
Confusion matrix: XGBoost (test, threshold 0.5)



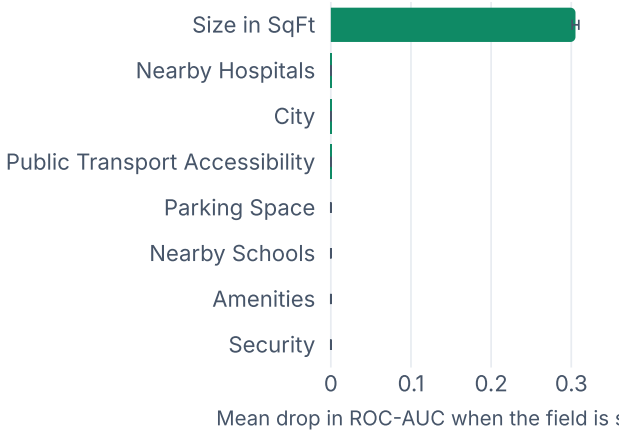
Champion test ECE 0.0031 is below the display threshold of 0.03, so probabilities are shown in the app.

Permutation importance

XGBoost, listing



Logistic Regression, attributes only



Drop in test ROC-AUC when one input is shuffled (5,000-row test sample, 3 repeats). It shows what the model uses, not what causes prices.

3. Valuation signal test

This is not a five-year forecast. The data has no future or historical prices, so a forecasting model cannot be trained or validated. The test asks whether today's price can be predicted from a property's attributes.

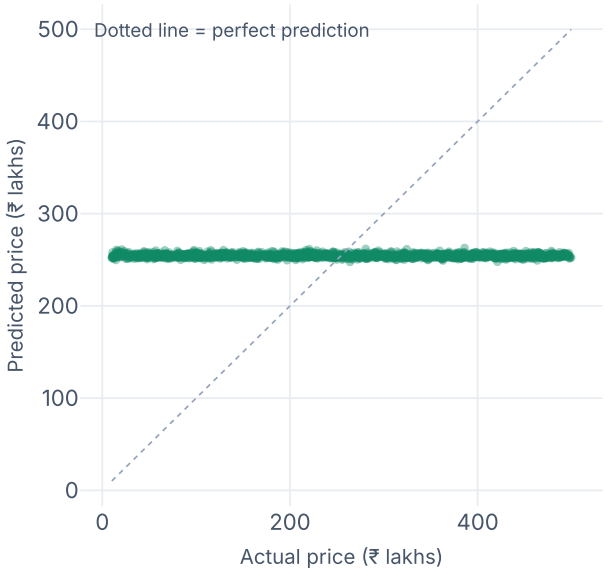
Regression on current price

Model	Val R^2	Test R^2	Test MAE (₹ L)	Test RMSE (₹ L)
Mean baseline	-0.0000	-0.0000	122.63	141.64
Ridge	-0.0003	-0.0005	122.65	141.68
Random Forest	-0.0002	-0.0002	122.64	141.66
XGBoost	-0.0037	-0.0040	122.79	141.93
Comparable-median rule	-0.7335	-0.7384	153.00	186.75

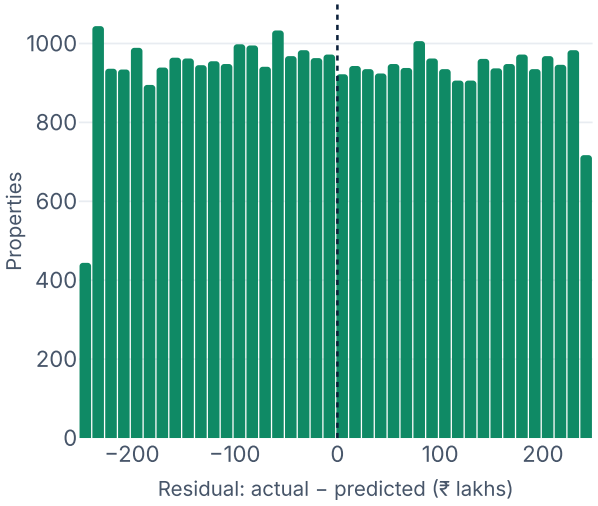
No learned model beats predicting the mean ($R^2 \approx 0$). The comparable-median rule (cohort median ₹/sqft × area) is worse than the mean because price is unrelated to area in this data. **There is no valuation signal to learn**, so the app does not present an estimated current value.

Diagnostics

Actual vs predicted: Random Forest



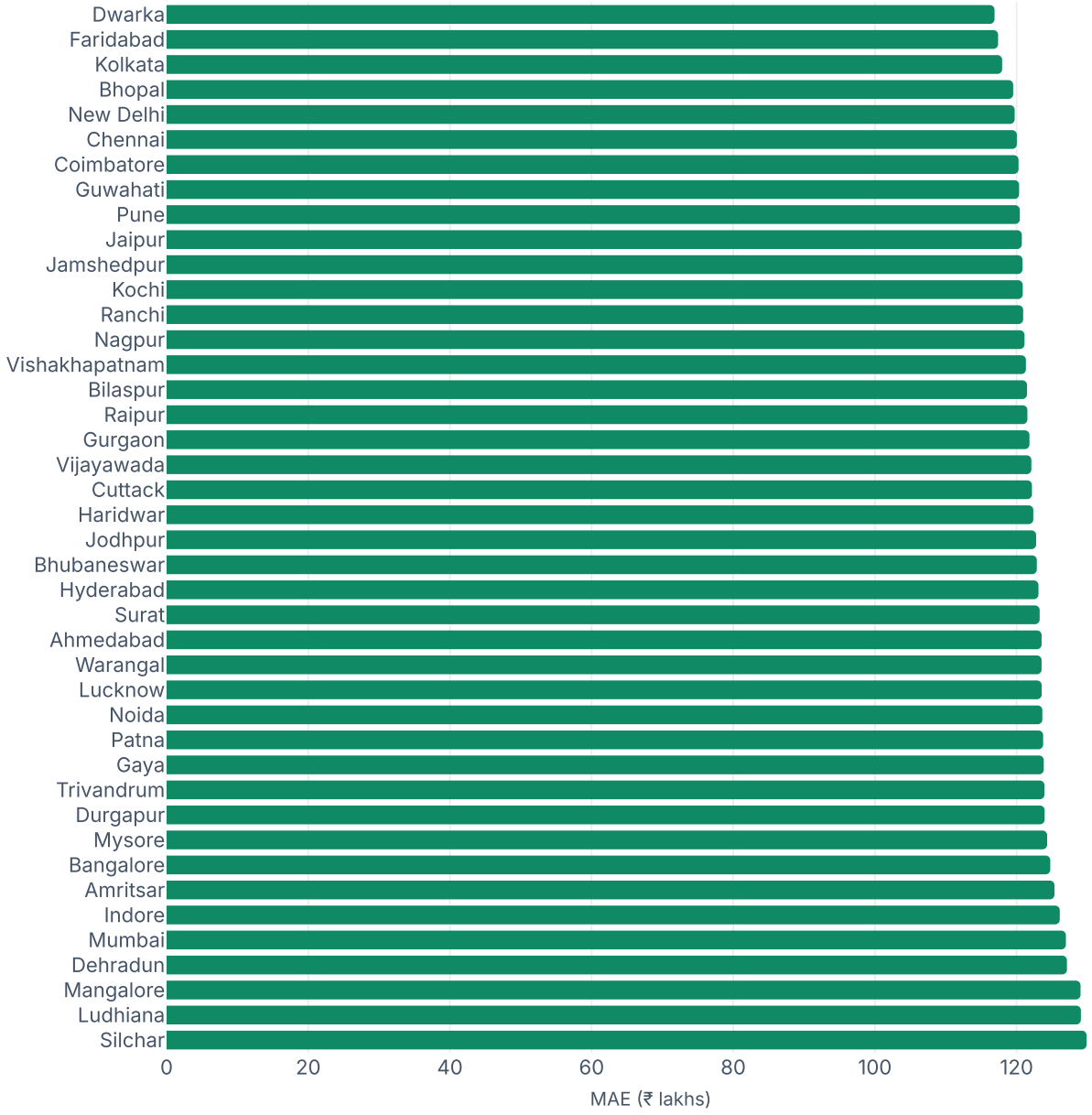
Residuals: Random Forest



Predictions cluster around the average price whatever the actual price: the signature of a target with no learnable relationship to the inputs.

Error by city

Mean absolute error by city (test)



MAE ranges only from ₹116.9 L (Dwarka) to ₹129.9 L (Silchar): error is flat across cities.

4. Five-year scenarios

Scenario engine (src/models/scenario.py)

Future value = current price × $(1 + r)^5$. The rates are assumptions, not predictions.

Scenario	Annual rate	Multiplier over 5 years
Conservative	2%	1.104×
Baseline	4%	1.217×
Optimistic	6%	1.338×

Baseline 4% sits between the latest RBI (3.6%) and NHB (5.0%) all-India annual readings. Conservative (2%) and Optimistic (6%) bracket it by ±2 points. These are single recent readings of national indices, not forecasts, and city-level growth varies widely (NHB Dec-2025 city readings ranged from −8.9% to +22.8%).

Sources

- RBI All-India House Price Index, Q1 2026-27 (released 24 Aug 2026)
- NHB RESIDEX, Q3 FY2025-26 (Oct–Dec 2025), 50-city HPI @ assessment prices

5. Experiment tracking (MLflow)

Cross-check against the tracking store

- 11/11 runs found in MLflow with matching test metrics.
- Registry: propwise-investment-classifier v1 (alias champion), status READY.

Classification experiment run IDs: champion `319810ac995d4a71b62082420803faa2`. Tracking store: `mlflow.db` (SQLite), artefacts in `mlartifacts/`.

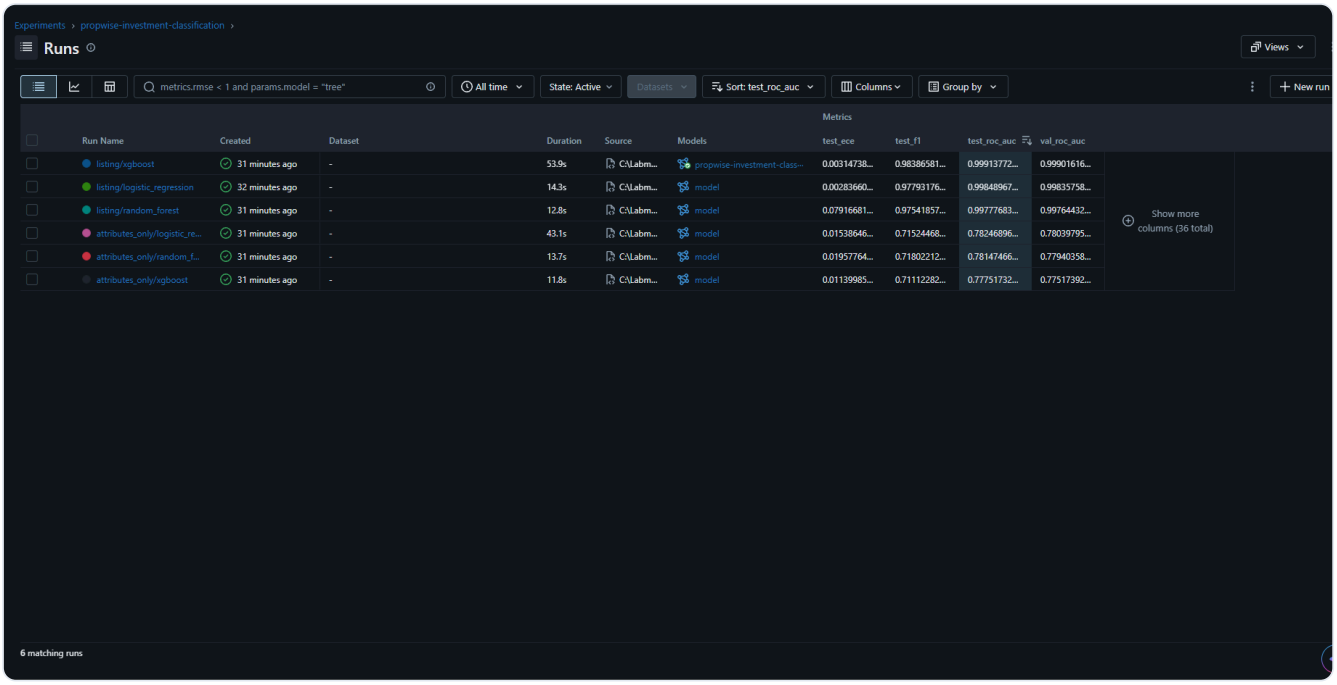


Figure 1. Classification experiment: six runs (two tracks × three models) with test metrics as columns.

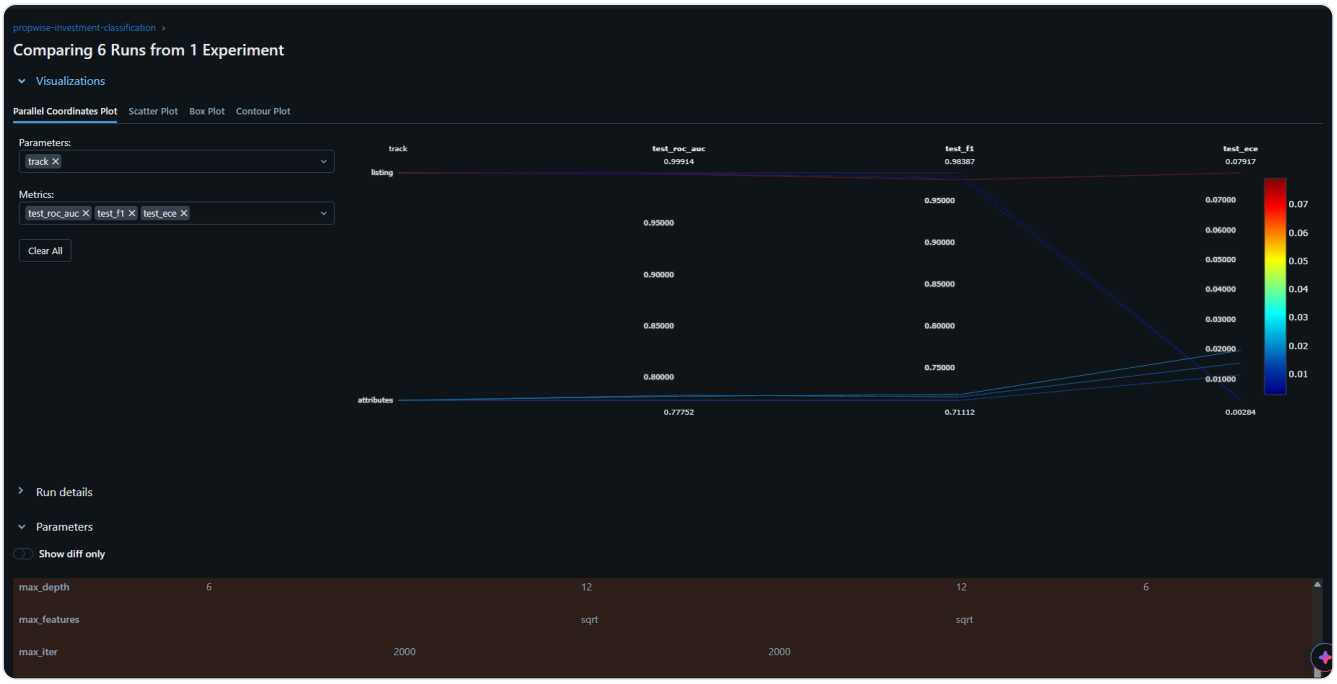


Figure 2. Run comparison: parallel coordinates across track, test ROC-AUC, F1 and ECE.

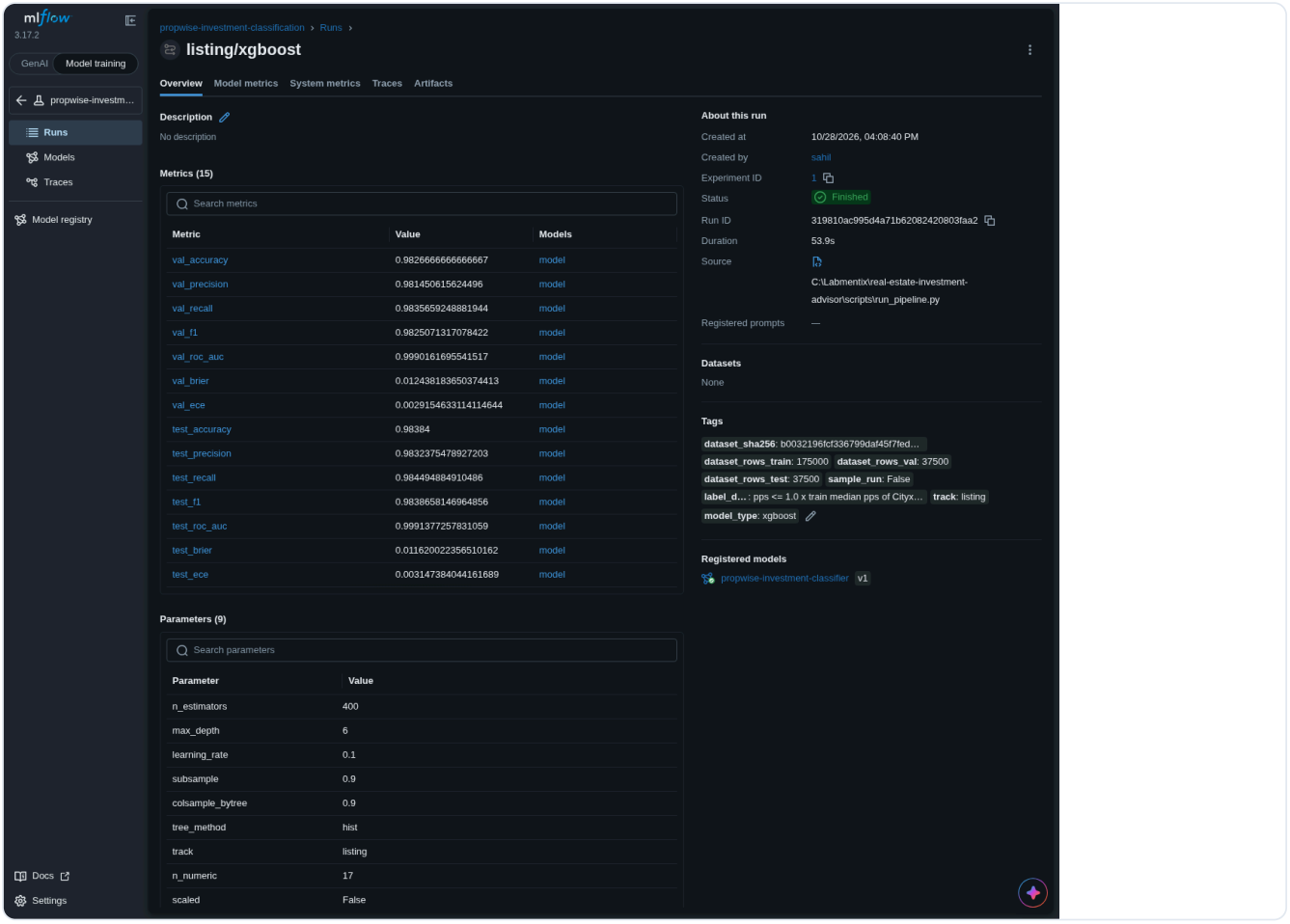


Figure 3. Champion run overview (listing / XGBoost).

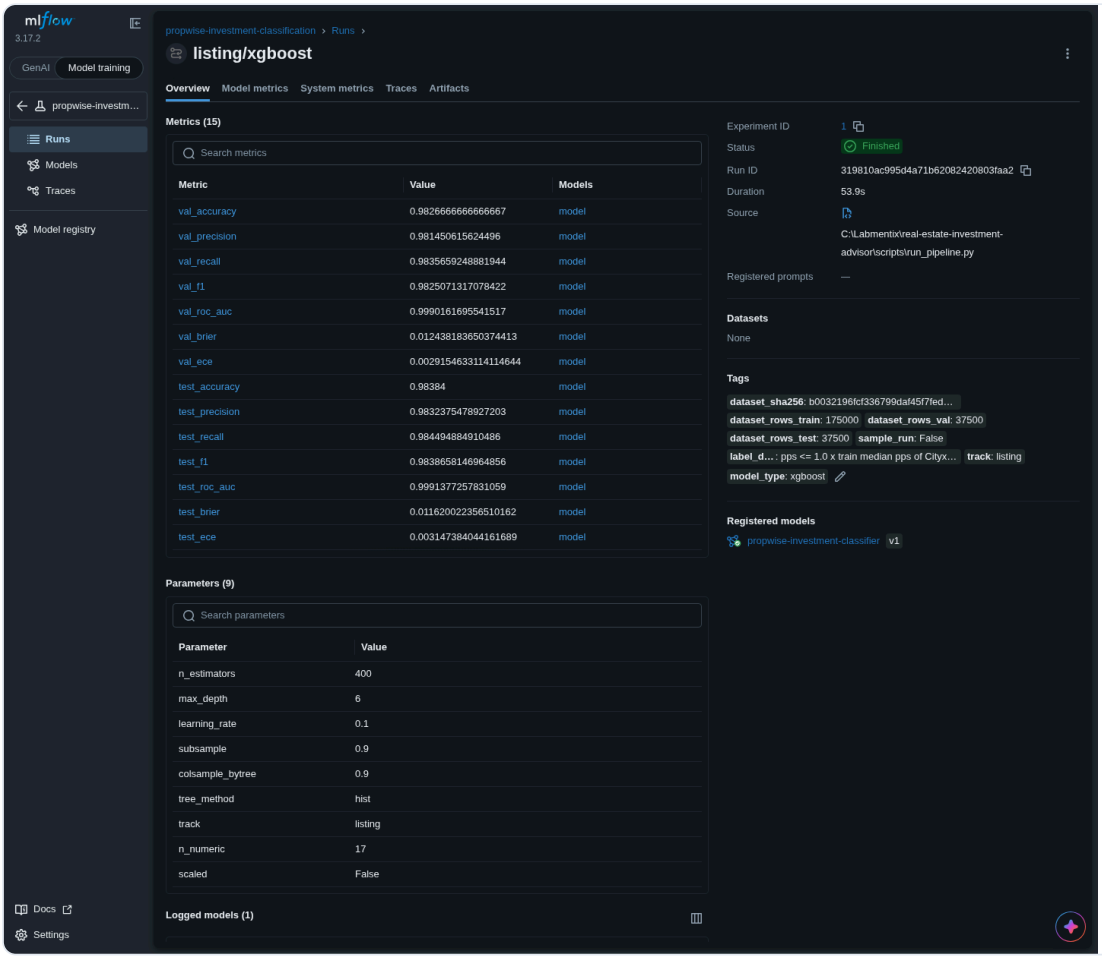


Figure 4. Champion run: logged metrics and parameters.

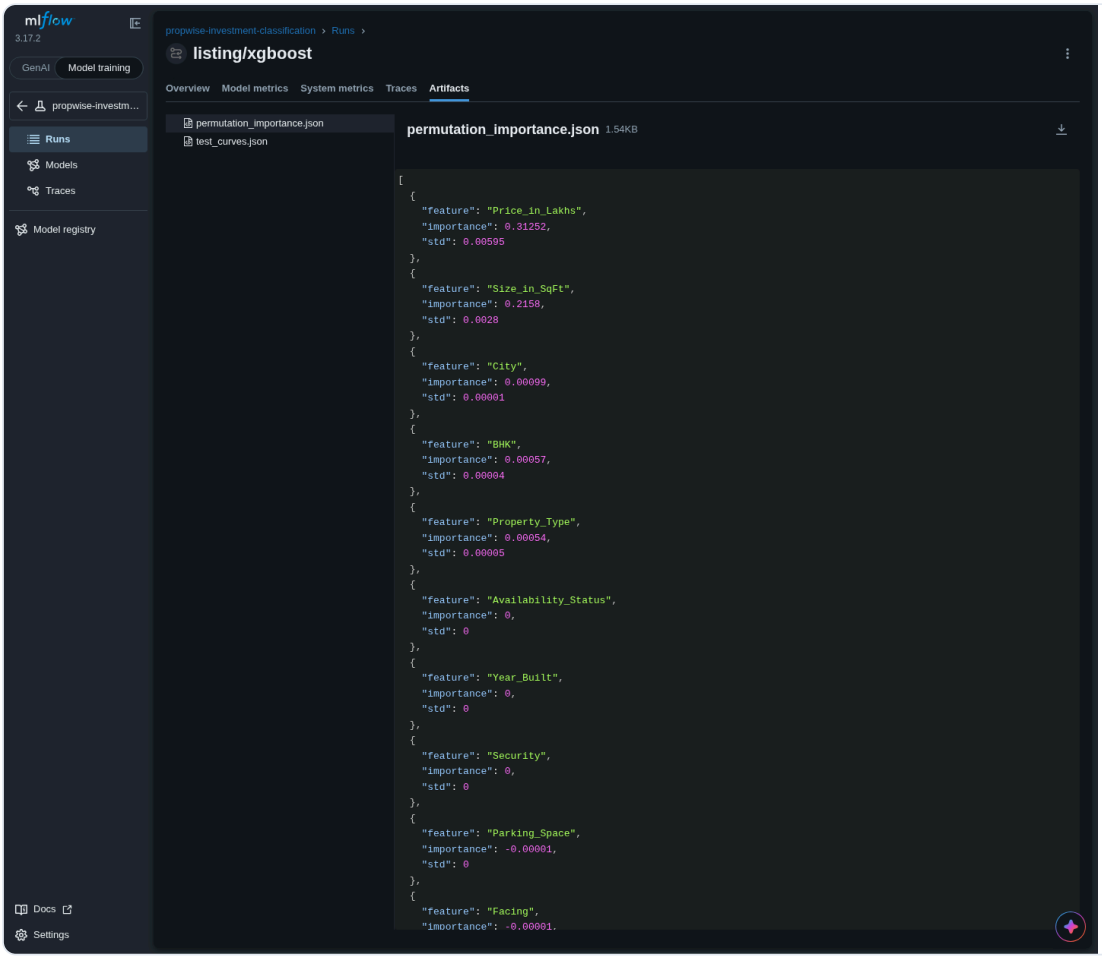


Figure 5. Champion run artefacts: the logged model and evaluation plots.

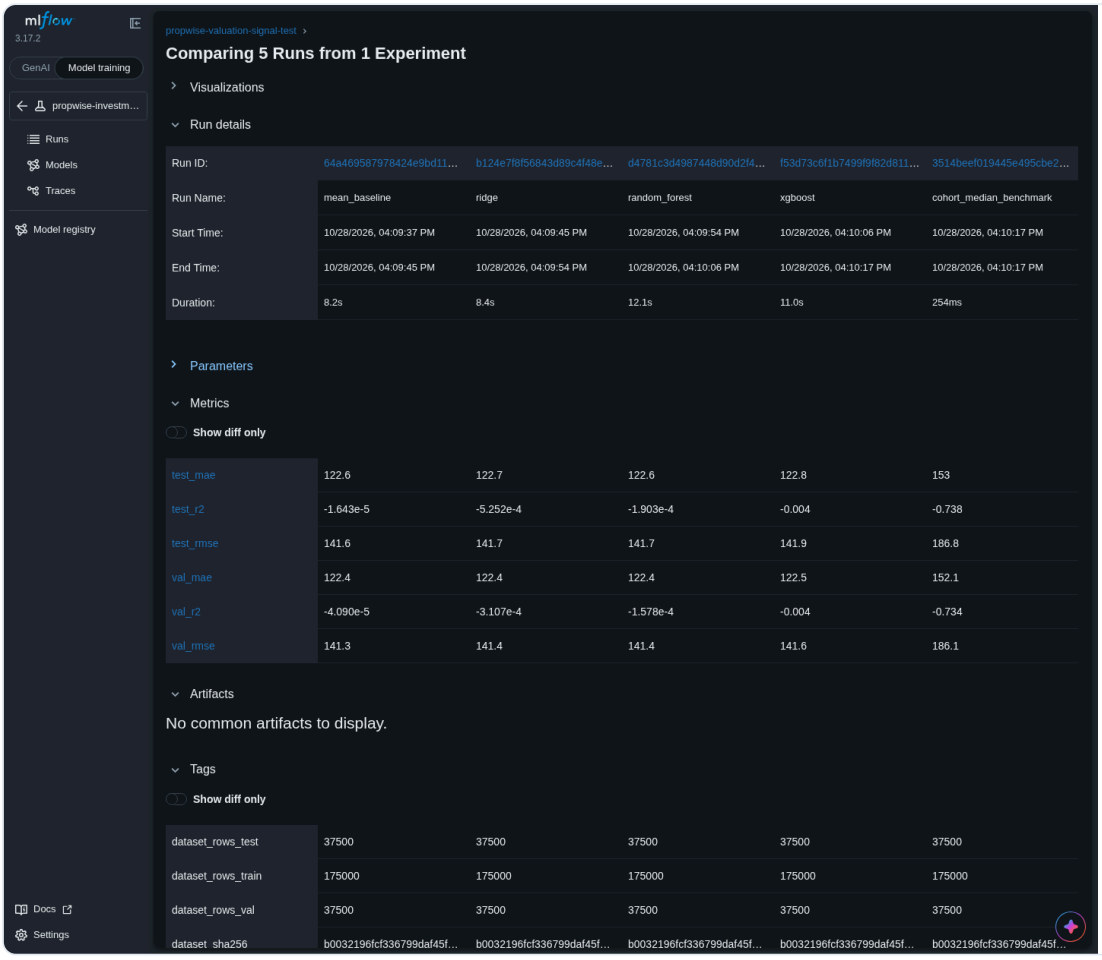


Figure 6. Valuation signal test experiment: every model at $R^2 \approx 0$.

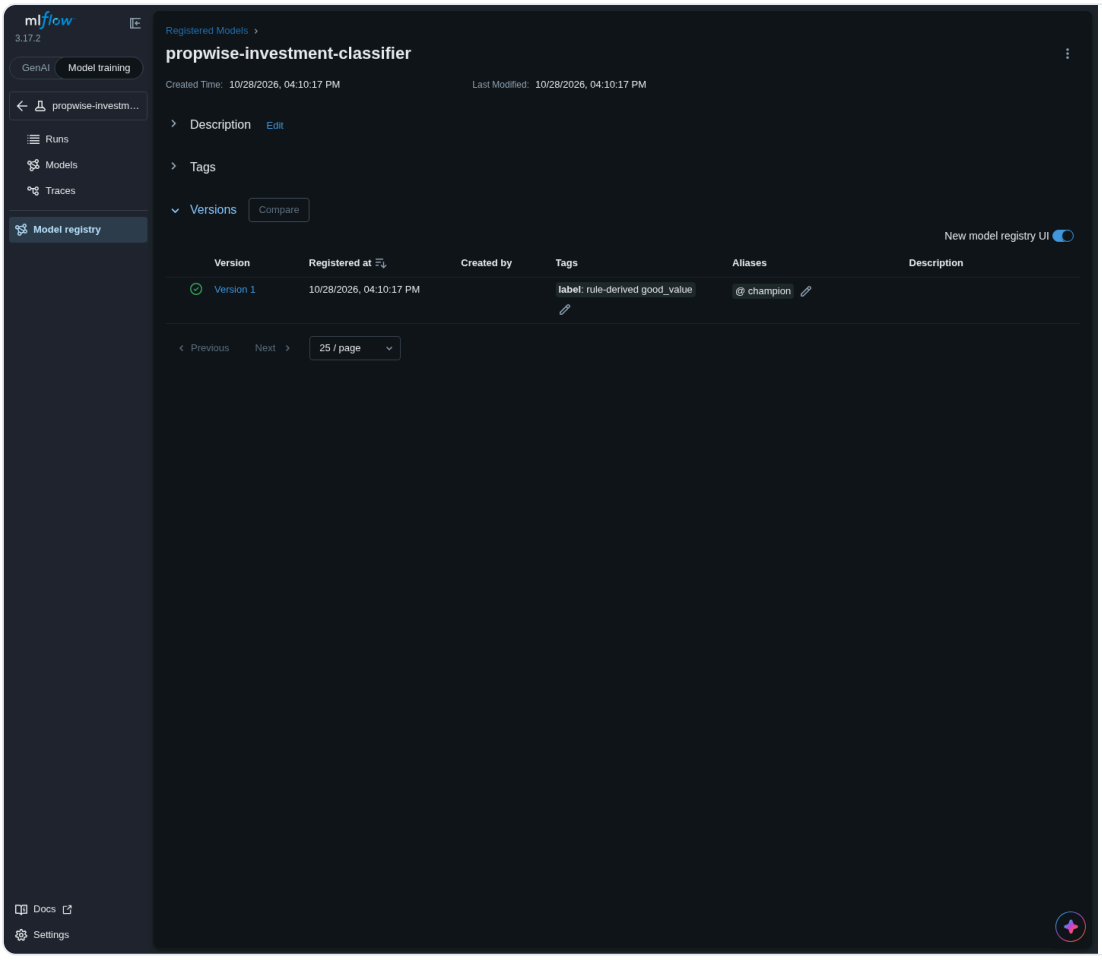


Figure 7. Model registry: propwise-investment-classifier, version 1 with alias @champion.

6. Reproducibility

Versions and commands

Package	Version
python	3.13.7
scikit-learn	1.9.1
xgboost	3.4.1
pandas	3.1.0
mlflow	3.17.2
numpy	2.5.4

Model bundle created 2026-10-28T10:40:17+00:00. Rebuild with `python -m scripts.train_models` then `python -m scripts.evaluate_models`.

Model scores describe this dataset only and are not investment advice.